AFF–1104

Sub. Code
4MCE1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science

APPLIED MATHEMATICS FOR COMPUTER SCIENCE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** the questions.

1. Let p stands for 'prices are high' and q for 'wages are rising'. Translate the following into verbal form : $p \vee \rceil q$.

2. Show that $Q \Rightarrow P \rightarrow Q$.

3. Obtain the principal disjunctive normal form of $P \leftrightarrow Q$.

4. Write the following is symbolic form (Assume that the universe consists of literally every thing) 'All men are grants'.

5. Define 'connected graph'.

6. Draw all trees with four vertices.

7. Define basic feasible solution.

8. While solving a LPP, maximize $z = cx$, such that $A_x = b$, $x \geq 0$ what indicate the unique optimal solution.

9. What is meant by an optimality test in a transportation problem?

10. What is an assignment problem?

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Show that $((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$ is a tautology.

Or

(b) Establish the following implication :

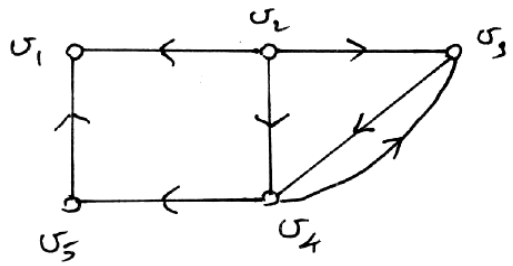
$(P \to (Q \to S)) \wedge (\neg R \vee P) \wedge Q \Rightarrow R \to S$.

12. (a) Obtain the principal conjunctive normal form of the formula : $(\neg P \to R) \wedge (Q \leftrightarrow P)$.

Or

(b) Let the universe consists of integers 1, 2, 4, 8, 16, 32. Let $P(x) : x$ is an even integer, $Q(x) : x$ is divisible by 8, $R(x, y) : x$ is divisible by y. Find the truth value of each of the following : $P(8)$, $P(16)$, $Q(4)$, $Q(32)$, $R(1,16)$, $R(16,1)$, $P(8) \wedge R(8,4)$.

13. (a) Find the adjacency matrix, incidence matrix for the following digraph.



Or

(b) Draw expression tree for the following expression $(a * b) - (c \div d) + e$.

14. (a) An animal feed company must produce 200 lbs of a mixture containing the ingredients x_1 and x_2, x_1 coasts Rs. 3 per lb and x_2 costs Rs. 8 per lb. Not more than 80 lbs. of x_1 can be used and minimum quantity to be used for x_2 is 60 lbs. Find how much of each ingredient should be used of the company works to minimise the cost. Formulate.

Or

(b) Use graphical method to solve the LPP :

Maximize $z = 2x_1 + 4x_2$,

Subject to $x_1 + 2x_2 \leq 5$, $x_1 + x_2 \leq 4$, $x_1, x_2 \geq 0$.

15. (a) Determine an initial basic feasible solution to the following transportation problem using North-West corner rule :

	D_1	D_2	D_3	D_4	Supply
O_1	5	3	6	2	19
O_2	4	7	9	1	37
O_3	3	4	7	5	34
Demand	16	18	31	25	

Or

(b) Solve the following assignment problem :

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{pmatrix} 2 & 10 & 9 & 7 \\ 15 & 4 & 14 & 8 \\ 13 & 14 & 16 & 11 \\ 4 & 15 & 13 & 9 \end{pmatrix} \end{array}$$

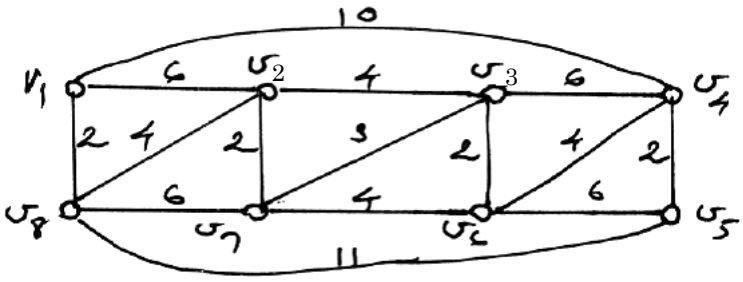
Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that $(\neg P \wedge (\neg Q \wedge R) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R)$.

17. Show that $(\forall x)(P(x) \vee Q(x)) \Rightarrow (\forall x)P(x) \vee (\exists x)Q(x)$.

18. Find the minimal spanning tree for the following graph :



19. Use simplex method to maximize $z = 2x_1 + 3x_2$ subject to the constraints $x_1 + x_2 \leq 4$, $-x_1 + x_2 \leq 1$, $x_1 + 2x_2 \leq 5$, $x_1, x_2 \geq 0$.

20. Solve the following transportation problem :

	To			
From	A	B	C	Available
I	50	30	220	1
II	90	45	170	3
III	250	200	50	4
Requirement	4	2	2	

AFF–1105

Sub. Code
4MCE1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science

DATA STRUCTURES AND ALGORITHMS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. What are the various types of data structure?

2. When the stack is overflow and underflow?

3. What is divide-and-conquer? Give an example.

4. What is the time complexity of binary search and selection sort?

5. Mention any two applications of greedy method?

6. What is meant by shortest path?

7. Mention any two differences between strongly connected and weakly connected graphs.

8. Define multistage graph.

9. State the principle of backtracking.

10. Mention any two uses of graph coloring.

Section B (5 × 5 = 25)

Answer **all** questions

11. (a) Write the algorithm for insertion and deletion of elements in a stack.

Or

(b) Write a short note on sets and disjoint sets.

12. (a) Write the algorithm for merge sort with its time complexity.

Or

(b) Discuss about selection sort with an example.

13. (a) Explain briefly about knapsack problem with an example.

Or

(b) Write about optimal storage on tapes briefly.

14. (a) Discuss the optimal binary search tree with its structure.

Or

(b) Write a short note on travelling salesperson problem.

15. (a) Write the algorithm for breadth first search.

Or

(b) Explain the construction of a biconnected graph with its structure.

Section C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Write the algorithm for insertion and deletion of elements in a queue.

17. Discuss the finding the maximum and minimum elements using straightforward and recursive methods briefly.

18. Explain about Prim's minimum spanning tree algorithm.

19. Discuss the multistage graph with a neat structure.

20. Write and explain the procedure for all solutions to the n-Queens problem.

AFF–1106

Sub. Code
4MCE1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science

ADVANCED JAVA PROGRAMMING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define JDBC.

2. Differentiate between SQL Exception and SQL Warning.

3. What are Sockets?

4. Expand RMI.

5. What is a jar file?

6. What do you mean by Persistence?

7. What is a Cookie?

8. Define HTTP.

9. What is Applet?

10. Define Panes.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain the connection class.

Or

(b) Explain a JDBC class.

12. (a) Discuss in short about InetAddress.

Or

(b) Explain about URL connection.

13. (a) Discuss about Bean Development Kit.

Or

(b) Explain the concept of Event and Methods.

14. (a) Explain the concept of session tracking.

Or

(b) Explain cookies.

15. (a) Describe the features of AWT.

Or

(b) Explain about various tables in java Applet.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss in detail about Meta Data Function with Examples.

17. Explain the concept of TCP/IP client and server Sockets.

18. Explain the various constrained properties with proper examples.

19. Discuss about Life cycles of Servlet.

20. Explain the Following in Java Applet

    (a) Button (b) Tree.

AFF–1107

Sub. Code
4MCE1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science

PRINCIPLES OF COMPILER DESIGN

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the possible error-recovery actions in Lexical analyzer?

2. Define FA.

3. Construct a parse tree of (a+b)*c for the grammar $E \rightarrow E + E \mid E * E \mid (E) \mid \text{id}$.

4. What is precedence parsing?

5. What is the intermediate code representation for the expression a or b and not c?

6. What are the methods of representing a syntax tree?

7. Write down the contents of a symbol table.

8. What is stack allocation?

9. What is an optimization compiler?

10. Define DAG.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is token? How token can be recognized using LEX tool?

Or

(b) Write an algorithm to construct an NFA from a given regular expression.

12. (a) Differentiate top-down parsing and bottom-up parsing.

Or

(b) Find the Context free grammar generating the language $L = \{a^n b^n ; n >= 1\}$ over the alphabet $\Sigma = \{a, b\}$.

13. (a) Explain syntax-directed translation schemes.

Or

(b) Discuss in briefly the translation of assignment statements.

14. (a) Discuss about storage allocation strategy in briefly.

Or

(b) Briefly explain about semantic errors.

15. (a) Discuss about the importance of loop optimization.

Or

(b) Write an algorithm to implement a simple code generator.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain in detail about the minimizing the number of states of a DFA and to recognize (a/b)* abb.

17. Construct the following grammar and construct a predictive parsing table

$E \rightarrow E + T \mid T$
$T \rightarrow T * F \mid F$
$F \rightarrow (E)/id$

18. How would you generate intermediate code for the flow of control statements? Explain with an example.

19. How to implement block? Explain.

20. Explain in detail about Register allocation and assignment in code generation.

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AFF–1108

Sub. Code
4MCE1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science

***Elective* — OBJECT ORIENTED ANALYSIS AND DESIGN**

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define OOAD.
2. Differentiate between object and classes.
3. What are nested state diagram?
4. Differentiate between events and states.
5. What is meant by iteration of analysis?
6. What is the goal of analysis?
7. What is automated Teller Machine?
8. Define data store.
9. List out the three models of object design.
10. Define optimization.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Define inheritance. What are the various forms of inheritance?

Or

(b) Describe about candidate keys.

12. (a) Discuss in detail about concurrence.

Or

(b) Explain about data flow diagrams with example.

13. (a) Discuss about functional modeling.

Or

(b) Explain the concept of dynamic modeling.

14. (a) Explain the concept of system design.

Or

(b) Explain about setting trade off priorities.

15. (a) Describe the features of Inheritance.

Or

(b) Explain about overview of object design.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss in detail about Metadata with example.

17. Explain the concept of advanced dynamic modeling.

18. Discuss briefly about object modeling with suitable illustrations.

19. Discuss the architecture of ATM system.

20. Explain the Following :

    (a) Object representation

    (b) Physical packaging.

AFF–1109

Sub. Code
4MCE1E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science

Elective — SOFTWARE ENGINEERING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define Software Testing.

2. What do you mean by Inspection Process?

3. What is Data flow Modeling?

4. State the use of Validation.

5. What is Project Scheduling?

6. Write any two functions of Detailed Scheduling.

7. What is meant by Problem Partitioning?

8. State a note on Verification.

9. What is White Box testing?

10. Define Testing Process.

Section B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write brief notes on Software Maintenance.

Or

(b) Describe briefly about prototyping model.

12. (a) Explain briefly about Requirement Specification.

Or

(b) Describe briefly about the Components of SRS.

13. (a) Write Short notes on Effort Estimation.

Or

(b) Describe briefly about Risk Management concepts.

14. (a) Explain briefly about Abstraction.

Or

(b) Describe briefly about top-down strategies.

15. (a) Explain briefly about Black Box testing.

Or

(b) Describe briefly about Boundary Value analysis.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Write brief notes on software Configuration Management process.

17. Explain the characteristics of an SRS.

18. Discuss in detail about COCOMO Model.

19. Explain about Design Notation and Specification.

20. Discuss in detail about the Data flow based Testing.

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AFF–1110

Sub. Code
4MCE2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science

COMPUTER SYSTEM ARCHITECTURE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the uses of Bus?

2. What are the different types of addressing modes?

3. Write any two logical shift micro operations.

4. How can the processor recognize the device requesting interrupt?

5. What are the basic requirements of memory management system?

6. Write any two advantages of microprogramming.

7. What is cache memory?

8. What is an IOP?

9. What do you mean by Arithmetic pipelining?

10. Define array processing.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write short notes on ALU.

Or

(b) Discuss briefly about stack organization.

12. (a) Explain briefly about instruction codes.

Or

(b) Discuss briefly about timing control.

13. (a) Briefly explain about micro program sequencer.

Or

(b) Write short notes on control memory.

14. (a) Discuss briefly about synchronous data transfer.

Or

(b) Discuss about the various priority interrupts.

15. (a) Briefly explain about the concepts of multiprocessing.

Or

(b) Discuss about the instruction pipelining.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain in detail about processor bus organization.

17. Discuss in detail about I/O interrupts.

18. Describe briefly about micro instruction formats.

19. Discuss in detail about virtual memory system.

20. Describe in detail about RISC architecture.

AFF–1111

Sub. Code
4MCE2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Second Semester

Computer Science

.NET TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is .Net frame work?

2. Define class library.

3. What is meant by Arrays?

4. What is meant by docking controls?

5. How to import namespace in ASP .Net?

6. How to use the AJAX file in ASP .Net?

7. Define class and objects.

8. Define abstraction.

9. What is meant by data objects?

10. Define reapters.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What is CTS? Explain any three .Net data type with relevant examples.

Or

(b) Write short note on .Net garbage collection.

12. (a) Write short notes on exception handling.

Or

(b) Explain the graphics and file handling.

13. (a) State the use of HTML TEXTAREA, TEXTANCHOR.

Or

(b) Discuss the different modes of session state to preserve the state information.

14. (a) Compare connection object and command object.

Or

(b) Write the various types of inheritance with suitable example.

15. (a) Write in detail about ASP .Net core object.

Or

(b) Illustrate the groups of ADO .Net objects and its functions.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain the architecture of CLR.

17. Explain the following :

    (a) Checkboxes.

    (b) Radio button.

    (c) List boxes.

    (d) Combo boxes. (2.5 × 4 = 10)

18. (a) Discuss the properties of a page class. (5)

    (b) Discuss about the various HTML server controls. (5)

19. Explain in details about different types of security models.

20. Write an ASP .Net code to fetch the data from SQL database table.

AFF–1112

Sub. Code
4MCE2C3

M.Sc. (CS) DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science

OPERATING SYSTEM

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the three main purposes of an operating system?
2. State the purpose of system calls.
3. List any two differences between user-level threads and kernel-level threads.
4. Differentiate between preemptive and nonpreemptive scheduling.
5. Is it possible to have a deadlock involving only one single process?
6. What is critical section problem?
7. Why page sizes are always powers of 2?
8. What is the cause of thrashing? How does the system detect thrashing?
9. Why must the bit map for file allocation be kept on mass storage rather than in main memory?
10. How do caches help improve performance?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write a brief note on Parallel systems.

Or

(b) Narrate about Storage Hierarchy.

12. (a) Explain briefly about Cooperating Processes.

Or

(b) Make a note on Threads.

13. (a) Discuss about Synchronization hardware.

Or

(b) Write a note on Recovery from Deadlock.

14. (a) Give the basic concepts about Paging.

Or

(b) Explain about Demand Paging.

15. (a) Discuss about Free Space Management.

Or

(b) Write short notes on Stable Storage Implementation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What are the system components of an operating system? Explain.

17. Discuss about any two CPU scheduling algorithms.

18. Explain the Banker's algorithm for deadlock avoidance.

19. Describe about Contiguous memory allocation.

20. What are files? Explain the access methods for files.

AFF–1115

Sub. Code
4MCE2E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science

Elective – COMPUTER GRAPHICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is computer art?

2. Write the merits of character generation.

3. What are the characteristics of good line drawing algorithm?

4. What is composite transformation?

5. Define view port.

6. Define work station.

7. Define scaling.

8. What is a 3D graphic package?

9. What is projection?

10. What is viewing parameters?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain Bresenham's line drawing algorithm in detail.

Or

(b) Discuss about the display processor.

12. (a) Explain about various area filling algorithm.

Or

(b) Explain 2D transformations with an example.

13. (a) Discuss about window to viewport transformation.

Or

(b) Briefly explain about interactive picture construction techniques.

14. (a) Explain 3D rotation about arbitrary axis.

Or

(b) Write short notes on 3D display techniques.

15. (a) Explain about projections with its types.

Or

(b) How object coordinate is converted into view plane coordinate? Explain.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Briefly explain about circle generating algorithm.

17. Discuss about the attributes of output primitives in detail.

18. What is segment? Explain about segment files and discuss its merits and demerits.

19. Explain about 3D transformations.

20. Discuss about hidden surface and hidden line removal algorithms.

AFF–1117

Sub. Code
4MCE2E5

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science

Elective — RELATIONAL DATA BASE MANAGEMENT SYSTEMS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write about the difference between DBMS and RDBMS.

2. What is meant by key? Discuss various types of keys.

3. Define the term index.

4. What is meant by hashing?

5. Write down the various operations on Relational Algebra.

6. What are characteristics and advantages of SQL?

7. Mention the ACID properties of transaction.

8. What do you mean by Database Recovery?

9. List the types of joins used in writing sub queries.

10. What are the various oracle database objects?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the different data models in RDBMS.

Or

(b) Discuss on hashing techniques.

12. (a) Discuss File organization based on dynamic hashing with Deferred Splitting.

Or

(b) Explain the B-Tree data structure.

13. (a) Explain the fundamental operations in Relational Algebra with examples.

Or

(b) Discuss the term decomposition. List its properties.

14. (a) What are the various concurrency control techniques? Explain.

Or

(b) What do you mean by locks? Explain read and write locks.

15. (a) How will you join multiple tables in a query? Explain.

Or

(b) Explain Embedded SQL commands in detail.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the creation and manipulation of indexed sequential file organization.

17. Explain in detail the secondary indexes with respect to B-tree.

18. Explain any two normal forms with examples.

19. Describe the statistical data base security.

20. Explain the following using oracle with examples :

    (a) Modifying tables

    (b) Renaming a table

    (c) Copying another table

    (d) Dropping a table.

AFF–1119

Sub. Code
4MCE3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

CRPTOGRAPHY AND NETWORK SECURITY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Computer Security.

2. What are the five ingredients of Symmetric Cipher?

3. How Block Cipher is different from Stream Cipher?

4. What is the purpose of S-box?

5. Mention the principle elements of a Public Key Cryptosystem.

6. Define one-way function.

7. What types of attacks are addressed by Message Authentication?

8. List out the properties of Digital Signature.

9. Why SSL is needed?

10. What are the five principal services provided by PGP?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the various categories of Security Services.

Or

(b) Compare and contrast Monoalphabetic and a Polyalphabetic Cipher.

12. (a) What is the difference between diffusion and confusion? Explain.

Or

(b) Discuss the importance of State Array.

13. (a) Differentiate Conventional Encryption and Public Key Encryption.

Or

(b) Explain the key ingredients of Public Key Directory.

14. (a) List and explain the characteristics that are needed in Secure Hash Function.

Or

(b) Discuss the disputes that can arise in the context of Message Authentication.

15. (a) Explain the important functions of S/MIME.

Or

(b) What are the applications and benefits of IPSec? Discuss.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Give a detailed note on Security Attacks with suitable examples.

17. Explain briefly about Data Encryption Standard (DES) with suitable sketch.

18. Describe the importance of Diffie-Hellman Key Exchange.

19. In what ways can a hash value be secured so as to provide Message Authentication? Justify your answer.

20. Discuss in detail about Encapsulating Security Payload with neat sketch.

AFF–1120

Sub. Code
4MCE3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

PROGRAMMING IN PHP

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List the strengths of PHP.

2. What are variables? How are variables declared in PHP?

3. What is the use of array( ) syntax in PHP?

4. What are global variables?

5. What are hidden fields?

6. State the difference between writing and appending to files.

7. What is meant by object serialization?

8. What is the use of join operation?

9. What is a session?

10. What are the applications of AJAX?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain the operators in PHP.

Or

(b) Explain the syntax of if and multiple if statements in PHP.

12. (a) Write a note on associative arrays.

Or

(b) Explain string functions in PHP.

13. (a) Describe client side validation.

Or

(b) Explain the PHP code for reading and writing text files.

14. (a) Explain object properties and methods.

Or

(b) Explain the syntax for setting and deleting cookies.

15. (a) Explain the methods and properties of XML HttpRequest object.

Or

(b) Describe the process of uploading and downloading files in PHP.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the basic syntax of PHP script and describe the datatypes and variables supported by PHP.

17. Explain the syntax for passing arguments to and returning values from user defined functions in PHP.

18. With an example program, explain form handling in PHP.

19. Explain the operations on a database with subquery and multiple queries.

20. Explain the PHP code for working with session variables.

AFF–1121

Sub. Code
4MCE3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

DATA MINING AND DATA WAREHOUSING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Knowledge Discovery in Databases? How does it relate to data mining?

2. Mention any two major issues in data mining.

3. What are operational databases?

4. How a database design is represented in OLAP systems?

5. Can a data mining system generate only interesting patterns? Justify.

6. How to generate association rules from frequent item sets?

7. Define Clustering.

8. List any four requirements of cluster analysis.

9. What is meant by spatial database? Mention its features.

10. What are the data mining tasks performed on a text database?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain various methods of data cleaning in detail.

Or

(b) Write short notes on data reduction.

12. (a) Summarize about On Line Analytical Mining.

Or

(b) Explain about Indexing OLAP data.

13. (a) Discuss briefly about Backpropagation.

Or

(b) Make a note on Rule Extraction from a Decision Tree.

14. (a) Write short notes on overview of basic clustering methods.

Or

(b) Narrate about Expectation-Maximization Algorithm.

15. (a) Write a note on Multimedia data mining.

Or

(b) Explain briefly about Text mining.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about data mining task primitives.

17. Discuss about Data Warehouse Architecture.

18. Compare and contrast Rough set approach with Fuzzy set approach.

19. Write a detailed note on any one partitioning methods.

20. Explain how data mining is useful in Science and Engineering.

AFF–1122

Sub. Code
4MCE3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

Elective – SOFT COMPUTING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define Computing.

2. What are Machine Learning basics?

3. What do you mean by the Genetic Algorithm?

4. State a note on Knowledge Acquisition.

5. What are Adaptive Networks?

6. Write any two advantages of Neural Networks.

7. What is meant by Fuzzy set?

8. What is the purpose of Fuzzy reasoning?

9. What is Neuro-Fuzzy Modeling?

10. Define Clustering.

Section B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Give a detailed on Soft Computing Constituents.

Or

(b) Write short note on Evolution of Computing.

12. (a) Explain briefly about the Applications of Genetic Algorithms.

Or

(b) Describe briefly the GA in Machine Learning.

13. (a) Write Short notes on Feed Forward Networks.

Or

(b) Explain briefly about Radial Basis Function Networks.

14. (a) Explain briefly the Operations on Fuzzy set.

Or

(b) Describe briefly the Fuzzy Expert Systems.

15. (a) Explain briefly about the Regression trees.

Or

(b) Describe briefly the Neuro-Fuzzy Control.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Give a detailed note on machine approach to knowledge acquisition.

17. Explain the Supervised Learning Neural Networks.

18. Discuss in detail about Fuzzy Relations.

19. Explain about Fuzzy Inference Systems.

20. Discuss in detail about the Data Clustering Algorithms.

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AFF–1124

Sub. Code
4MCE3E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

Elective — MULTIMEDIA SYSTEM

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define multimedia.
2. What are the various hardware components of multimedia?
3. What do you mean by digital video?
4. Define animation.
5. List down the various audio file formats.
6. Differentiate JPEG and MPEG.
7. What is MIME?
8. Write down the applications of multimedia.
9. What is virtual reality?
10. What are the various modes of interaction in virtual reality?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write short notes on multimedia networking.

Or

(b) Discuss briefly about multimedia standards.

12. (a) Explain briefly about triggering.

Or

(b) Discuss briefly about various CD formats.

13. (a) Explain briefly about MIDI sequencing video technology.

Or

(b) Write short notes on video compression standards.

14. (a) Discuss briefly about video teleconferencing.

Or

(b) Write brief note on HTML and XML.

15. (a) Briefly explain about the concepts of generic virtual reality system.

Or

(b) Discuss about VR hardware.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain in detail about multimedia software and multimedia PC.

17. Explain the following : $(4 \times 2.5 = 10)$

    (a) Shadowing

    (b) Cloning

    (c) Positioning

    (d) Sizing.

18. Discuss the steps involved in digital video recording and video clip making.

19. Explain briefly about multimedia presentation and authoring design.

20. Discuss briefly about sensor hardware and acoustic hardware.

AFF–1126

Sub. Code
4MCE3E5

M.Sc. (CS) DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

Elective : CLOUD COMPUTING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is cloud computing?

2. List any two properties of cloud computing.

3. Who get benefited from cloud computing?

4. Mention any two demerits of cloud development.

5. How cloud computing helpful in event schedules and management?

6. Can you share family photos using Cloud Computing?

7. What is the use of Event Calendar?

8. List any four Project Management applications.

9. Mention the use of groupware.

10. Name some of the popular social networks.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What are the advantages of Cloud computing?

Or

(b) Discuss about Cloud Services.

12. (a) Explain the concept of On-Demand computing.

Or

(b) Narrate about Google App Engine.

13. (a) Discuss about Centralizing Email communications.

Or

(b) Write short notes on Collaborating on To-Do Lists.

14. (a) Explain briefly about understanding event management applications.

Or

(b) How Web-based word processing works? Discuss.

15. (a) Summarize about any two largest Web mail services.

Or

(b) Write a note on evaluating blogs for collaboration.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about the history of Cloud computing.

17. Discuss the Pros and Cons of Cloud Service Development.

18. Write a detailed note on Collaborating on Group Projects and Events.

19. Elaborate about Collaborating on databases.

20. Explain in detail about evaluating Web conferencing tools.

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AFF–1127

Sub. Code
4MCE3E6

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

Elective — WAP & XML

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the disadvantages of using layered architecture?

2. Define WAP resources.

3. What is a Markup language?

4. What are the attributes that are supported by all WML elements?

5. What tags are used to define hotspots in WML?

6. What are transcoders?

7. Define XML.

8. List out the applications of XML.

9. Define empty tags.

10. State the need for XSL.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write a brief note on origin of WAP.

Or

(b) Discuss about the various wireless middleware solutions.

12. (a) Explain the functionality of a gateway.

Or

(b) Write short note on WML card.

13. (a) Describe the lexical structure in detail.

Or

(b) What is WML authoring? Briefly explain.

14. (a) Explain the various technologies used to develop XML document.

Or

(b) List the advantages of XML format.

15. (a) Difference between attributes versus elements in XSL.

Or

(b) Write note on legacy character sets.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the WAP development toolkit.

17. Explain about the Design Principles of WAP Architecture.

18. Explain in detail about WML document model.

19. Explain XML document type declaration.

20. Explain the procedure to write XML Unicode.

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